Problem 1

(20pt) When proving the NP-completeness of SAT, we converted the NAND gate to a boolean formula. Write down this boolean formula in the conjuctive normal form. More concretely, write down a boolean formula $f(x_1, x_2, x_3)$ so that

$$f(x_1,x_2,x_3) = \begin{cases} \text{True} & \text{if } \neg(x_1 \wedge x_2) = x_3, \\ \text{False} & \text{else.} \end{cases}$$

AND:

x_1	x_2	x_3 (AND)
0	0	0
0	1	0
1	0	0
1	1	1

NAND

x_1	x_2	x_3 (NAND)
0	0	1
0	1	1
1	0	1
1	1	0

DNF of AND:

$$(\overline{x_1} \wedge \overline{x_2} \wedge \overline{x_3}) \vee (\overline{x_1} \wedge x_2 \wedge \overline{x_3}) \vee (x_1 \wedge \overline{x_2} \wedge \overline{x_3}) \vee (x_1 \wedge x_2 \wedge x_3)$$

$\Rightarrow$ CNF of NAND:

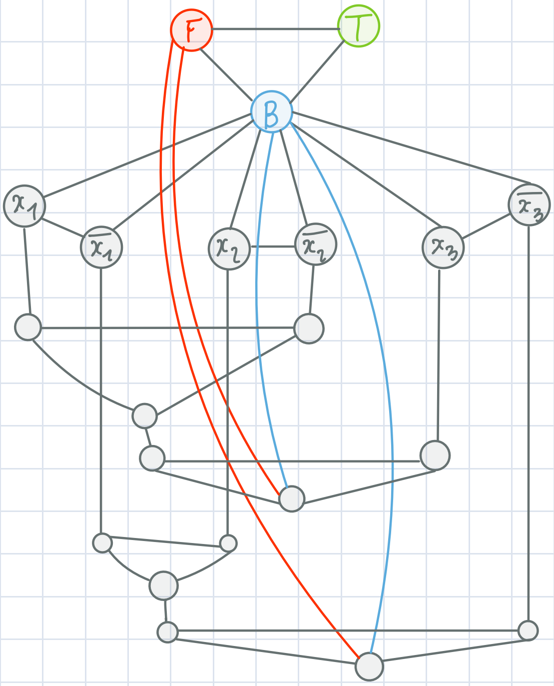
$$\neg((\overline{x_1} \wedge \overline{x_2} \wedge \overline{x_3}) \vee (\overline{x_1} \wedge x_2 \wedge \overline{x_3}) \vee (x_1 \wedge \overline{x_2} \wedge \overline{x_3}) \vee (x_1 \wedge x_2 \wedge x_3)) \quad \text{(De Morgan's Law)}$$

$$\equiv (x_1 \vee x_2 \vee x_3) \wedge (x_1 \vee \overline{x_2} \vee x_3) \wedge (\overline{x_1} \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_3})$$

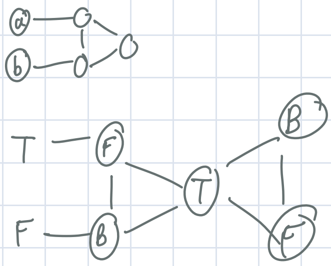
Problem 2

(20pt) Recall the procedure described in class for reducing an instance of 3SAT to the 3-coloring problem. Following the same procedure, convert the following boolean formula $f(x_1, x_2, x_3)$ to a graph, so that a decider for the 3-coloring problem can be used to decide the 3SAT instance.

$$f(x_1, x_2, x_3) = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee \neg x_3).$$



OR:



Problem 3

(20pt) A *vertex cover* of a graph G is a subset of vertices where every edge of G touches one of these vertices.

$$\text{VERTEX-COVER} = \{\langle G, k\rangle | G \text{ is an undirected graph that has a vertex cover with } k \text{ nodes.}\}$$

Prove that VERTEX-COVER is NP-complete by reducing CLIQUE to VERTEX-COVER.
Hint: Suppose you have a graph G with a k-clique, denoted as C. In the complement graph of G, there is a vertex cover that is closely related to C. Once you figure out this vertex cover, the reduction would be immediate.

Claim 1: VERTEX-COVER $\in$ NP

Let M be a DTM, $G = (V,E)$, $V' \subseteq V$ (V' is a given vertex cover)

M = "On input $\langle G, V' \rangle$:

1) count $= 0$
2) For every vertex $v \in V'$:
   Remove all edges adjacent to v
   increment count by 1.
3) If count $= |V'|$ and E is empty, accept
   Else, reject.

Claim 2: CLIQUE $\leq_p$ VERTEX-COVER.

Let $G=(V,E)$ a graph with a k-clique, denoted as C. ($|V| = n$)

Create the following graph $\overline{G}$:

$V' = V$

$\overline{E}$ = If there is an edge between 2 vertices in G, remove them in $\overline{G}$
Else, create an edge in $\overline{G}$ if there hasn't existed an edge between 2 vertices in G.

We will see that the remaining vertex with adjacent edges in $\overline{G}$ is the vertex cover for $\overline{G}$.

In fact, the remaining vertices with adjacent edges are $n-k$.

Now, constructing $\overline{G}$ takes $O(n^2)$ time at most since we need to check each pair of vertices G, and there exists approximately $O(n^2)$ pairs of vertices in G.

Proof for correctness:

G has a k-clique $\Longleftrightarrow$ $\overline{G}$ has a vertex cover of size $n-k$.

- G has a k-clique $\Longrightarrow$ $\overline{G}$ has a vertex cover of size $|V|-k$.

  Let G has a k-clique $C(V', E')$. Let $X = V \setminus V'$

  (WTS: X is a vertex cover of $\overline{G}$ of size $n-k$)

  Suppose we have an edge $(u,v) \in \overline{E}$ of $\overline{G}$, where both u and $v \in V'$ of C.

  Since C is a clique in G, there must be an edge $(u,v) \in E$ of G.

  $\Rightarrow$ Contradiction with the fact that such edge should not exist in $\overline{E}$ (by definition of $\overline{G}$)

  $\Rightarrow \forall (u,v) \in \overline{E}$, at least one of u or v must belong to X.

  $\Rightarrow X$ is a vertex cover of $\overline{G}$ and its size is $|V|-k$.

- $\overline{G}$ has a vertex cover of size $n-k \Rightarrow G$ has a k-clique.

  Let $\overline{G}$ has a vertex cover X of size $n-k$. Let $C = V \setminus X$

  (WTS: C is a clique in G of size k)

  Take 2 distinct vertices $u, v \in C$.

  Suppose, (u,v) is an edge in $\overline{G}$. However, since X is a vertex cover, (u,v) must have at least one endpoint in X.

  $\Rightarrow$ Contradiction since u and v are in C which means they are not in X (by definition of C)

By claim 1 and 2, VERTEX-COVER is NP-complete.

Problem 4

Show that PSPACE is closed under the following operations.

(a) (10pt) Union

(b) (10pt) Kleene star

(a) Union :

Let $A, B \in PSPACE$.

Then, there exists DTM D_A and D_B as deciders using polynomial space.

Now, construct D_C as follow:

D_C = "On input w

Run D_A on w. If D_A accepts, accepts

Else, run D_B on w → D_B accepts, accepts

D_B rejects, rejects.

We first copy w onto the tape, which takes $O(n)$ space.

By definitions, D_A takes at most $O(n^k)$ space for some $k \geq 0$ and D_B takes at most $O(n^l)$ space for some $l \geq 0$ on all input of length n.

Therefore, D_c takes $O(n^k + n^l + n)$ space, which is still polynomial.

$\Rightarrow D_C \in PSPACE$.

$\Rightarrow$ PSPACE is closed under union.

(b) Kleene star :

Suppose $L \in PSPACE$

Then there exists a decider TM_L that runs using polynomial space $p(n)$ for input $n \in L$.

Now, construct L^* as follow:

"On input $w = w_1 w_2 \ldots w_n$

1. Create a boolean array A with a size $|n| + 1$ and initializes all elements to be false.
   ($A[i]$ = true iff $w_1 \ldots w_j$, each $w_k \in L$)
2. Let $A[0]$ = true (since empty string is in L^*)
3. For $i = 1$ to n
   For $j = 0$ to $i-1$:
   $A[i] = A[j]$ and $TM_L(w_{j+1} \ldots w_i)$
4. If $A[n]$ = true, accept
   Else, rejects."

We use $O(n)$ to store the input

For step 1, 2 and 4, we use $O(n)$ space in the input size.

Step 3 have 2 loops and each iteration we run TM_L, but we don't need to store immediate result at once, but we can reuse the space that TM_L use $\Rightarrow O(p(n))$

Moreover, the indices used for the loops only take $O(\log n)$ space to store.

$\Rightarrow$ Total space : $O(\text{array}) + O(\text{indices}) + O(\text{Space for } TM_L)$

$= O(n) + O(\log n) + O(p(n))$

$= O(p(n))$ as TM_L scale with the input.

But TM_L uses as most poly-space by definition.

$\Rightarrow L^*$ uses as most poly-space

$\Rightarrow$ PSPACE is closes under Kleene star.

Problem 5

(20pt) Consider the set of balanced parantheses, denoted as B here. Show that $B \in L$.
* For instance,)($\notin B$ but (()()) $\in B$.

counter = 0

1. If) → reject)
2. If (, count += 1, move right.
   If see empty tape, reject
3. If) , count −= 1, move right
4. If (, go to 2.
   If), go to 3.
   If b, go to 5.
5. If count = 0, accept
   Else, reject

Since storing count only use $O(\log n)$ as count is just an index
$\Rightarrow B \in L$.